Reg. Number:	

VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Continuous Assessment Test (CAT) – I - AUGUST 2025

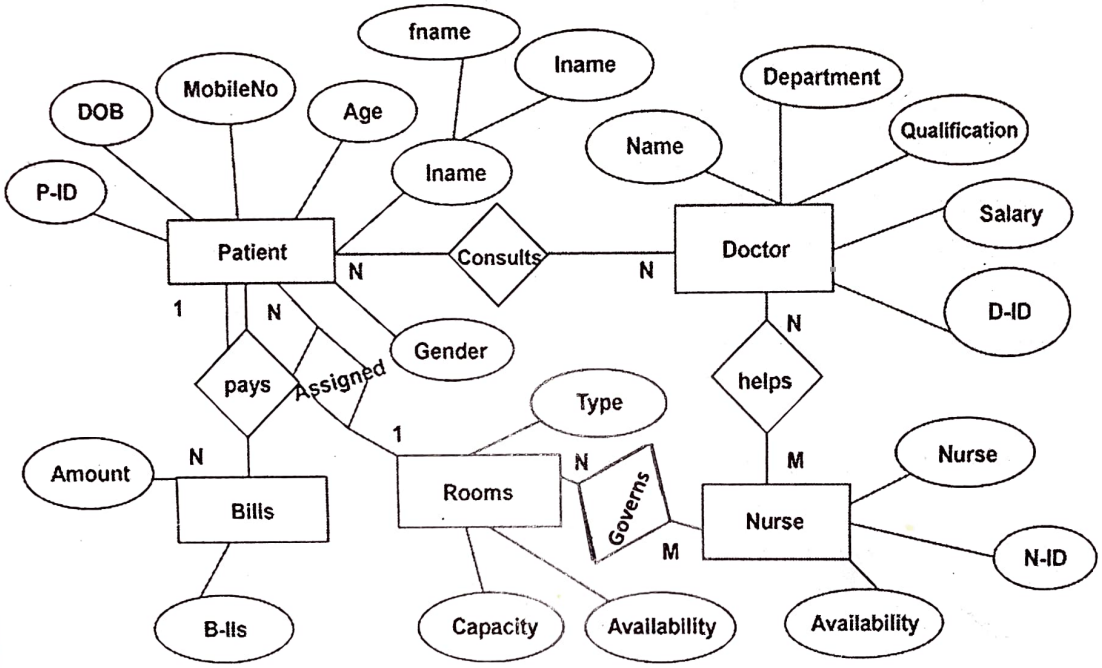
Programme	:	B.Tech	Semester	:	Fall Semester 2025-26
Course Code & Course Title	:	BCSE302L Database Systems	Class Number	:	CH2025260100635 CH2025260100636 CH2025260100637 CH2025260100638 CH2025260100642 CH2025260100643 CH2025260100645 CH2025260100646
Faculty	:	Dr. SIVARAMAKRISHNAN N, Dr. VIJAYAKUMAR K P Dr. JANI ANBARASI L, Dr. ASHOKA RAJAN R Dr. BHUVANESWARI, Dr. APPALARAJU MURALIDHAR, Dr. LENINISHA SHANMUGAM Dr. GAYATHRI DEVI S	Slot	:	B2+TB2
Duration	:	90 Minutes	Max. Mark		50 Marks

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- Use statistical tables supplied from the exam cell as necessary
- Use graph sheets supplied from the exam cell as necessary
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks
1	a)	Mr. Ram operates a grocery store and seeks an effective way to record and analyse purchase data, sales transactions, and customer information. He also requires the ability to generate regular reports daily, weekly, and monthly to monitor the store's performance. i. Should Mr. Ram use a conventional file-based system or implement a relational database using a DBMS? Justify your recommendation by comparing both approaches. (3 marks) ii. Explain the role of the three-schema architecture in structuring his database system, emphasizing how each level contributes to the system's usability and maintainability with neat sketch. (7 marks)	10
2	a)	A Multi-Speciality Hospital requires a database to manage patients (personal, contact, medical history, insurance), doctors (specialization, qualifications, contact, schedule), appointments (patient–doctor bookings, date, time, department), billing (consultation, treatment cost, payment status), and lab reports (linked to patients, including scanned/PDF files). The system must support many-to-many relationships between patients and doctors, ensure data integrity, avoid duplicates, enable fast queries (e.g., unpaid bills last month, Dr. Meera's appointments tomorrow), and store both structured data and unstructured lab reports. (i) Illustrate types of data models used in DBMS. (4 Marks) (ii) Select one data model for the hospital database, describe it briefly, and justify any 3 reasons for your choice. Explain the concept of data independence in DBMS for the above scenario with its types. (6 Marks)	10

3	a)	Consider the following Database schemas: Department (DeptID, DeptName ,Location) Professor (ProfID, ProfName, DeptID, HireDate) Course (CourseID, CourseName, Credits, DeptID, ProfID) Write SQL queries for the following: Write SQL queries to create the tables along with foreign key constraints where it is required ii) Add a constraint so that HireDate cannot be before '2000-01-01'. iii) Alter the Course table to add a column Semester of type Varchar(20) with default value 'Fall'. iv) Modify the Professor table to increase ProfName column length to 100 characters. v) Number of Credits (1 to 4) should be verified with suitable check constraint.	10
4	a)	Consider the following ER diagram for the Library Management System  Map the given ER diagram into a relational schema step by step with all possible constraints.	10
5	a)	Consider the following functional dependencies (FDs) For the relation schema R:(A,B,C,D,E,F,G,H,I) A → BCDE F → AGI GF → H CA → E F → H GI → H i) Find the canonical cover for the given set of functional dependencies. (8 marks) ii) Identify all the possible candidate keys(2 marks)	10

**********All the best **********